SIMATS ENGINEERING

CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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Register Number	**192525002**

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

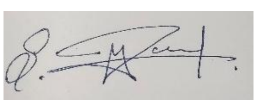
Question Count: 5

Code of Conduct

I **MAGESH S (192525002)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

Comparison :

Criterion	Type-1 Hypervisor (Bare-Metal)	Type-2 Hypervisor (Hosted)
Architecture	Runs directly on physical hardware (bare-metal); no host OS layer.	Runs as an application on top of a host operating system.
Performance	Near-native performance; direct hardware access via thin hypervisor kernel; low I/O overhead.	Extra translation layer through host OS adds latency and CPU overhead; noticeable under heavy load.
Security	Smaller attack surface (no general-purpose host OS to patch); strong VM isolation enforced at hardware layer; supports hardware-assisted security (Intel VT-x/VT-d, TPM).	Inherits host OS vulnerabilities; a compromised host OS can compromise every guest VM; larger attack surface.
Manageability	Centralized enterprise management (vCenter, SCVMM), live migration, HA/DRS, clustering — built for datacenter scale.	Simple to install and use for individual desktops/dev-test; lacks enterprise clustering and centralized fleet management.
Resource utilization	Efficient resource pooling and scheduling across many VMs on shared hardware.	Host OS consumes CPU/RAM before any VM workload even starts.
Typical use case	Production servers, datacenters, cloud infrastructure (ESXi, Hyper-V, KVM, Xen).	Developer sandboxes, testing, desktop virtualization (VMware Workstation, VirtualBox).

Justification

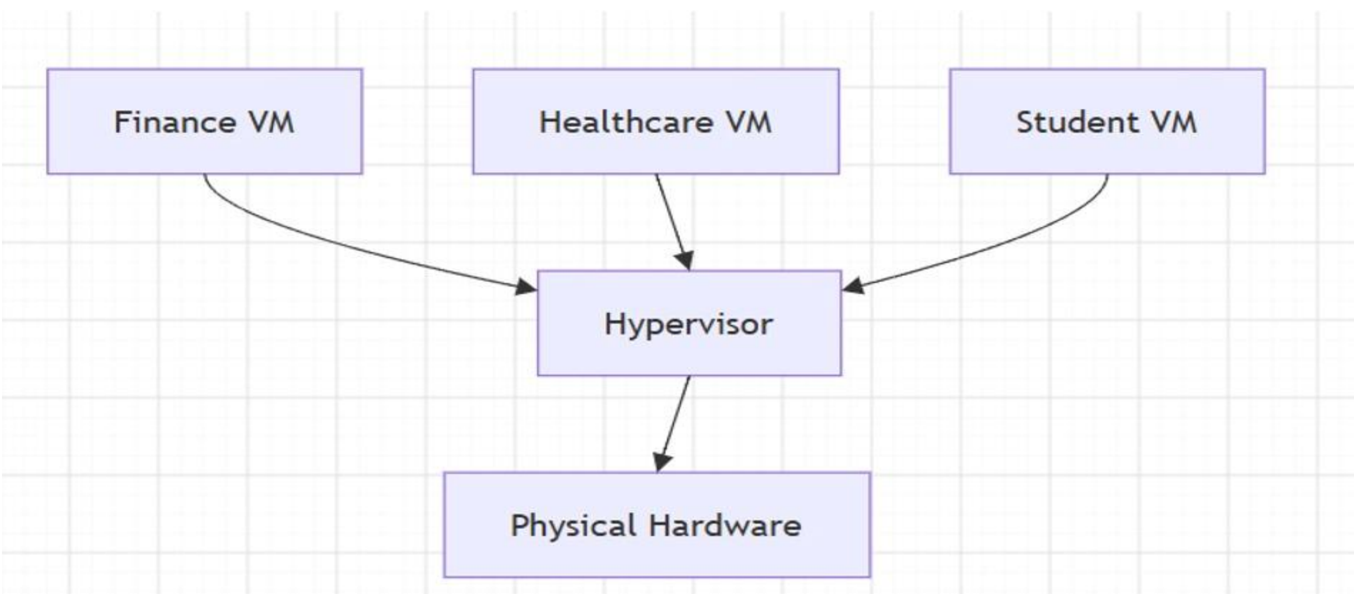
In case of a datacenter that requires to run mission-critical applications, Type-1 hypervisor is the more preferred one. For a mission-critical application to be executed successfully in a virtualized environment, there is need for deterministic low latency performance, which can only be provided by Type-1 hypervisor due to its capability of direct hardware access, there is need for tenant isolation and minimal attack surface, which is not possible with Type-2 hypervisor due to the vulnerability of its underlying host OS, and there is need for enterprise level manageability of the virtualized environment.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

How virtualization architecture enforces isolation

Hardware-enforced isolation: The use of CPU virtualization (Intel VT-x/AMD-V) features in the hypervisor, along with an IOMMU (Intel VT-d), creates a unique protected address space for every VM, meaning that one tenant cannot access another's memory.

- Allocation constraints: CPU shares/reservation and memory limits guarantee that one tenant does not consume all the resources of the physical machine, and thus the "noisy neighbor" phenomenon is avoided.
- Network micro-segmentation: VLAN/VXLAN tunneling and tenant-specific security groups ensure that finance, healthcare, and student portal network traffic flows through logically separate networks, despite the traffic being transported over the same physical switches.
- Storage isolation: Virtual disk images are isolated logically into distinct volumes, sometimes even with unique encryption keys per tenant, which ensures strict resource access within the storage layer.
- Management-plane RBAC: role-based access control on the hypervisor/orchestration layer guarantees that the administrators only access the resources of the respective tenant.



Two main attack surfaces

- Hypervisor/VM-escape attack surface: a weakness in the hypervisor would allow the attacker to take advantage of compromising one tenant VM (such as student portal) to escape from this VM sandbox to get access to the memory or other resources of another VM such as finance or healthcare VM on the same host machine.
- Common/shared network and management plane attack surface: The virtual switch, SDN controller and orchestration/api layer are common to all tenants; configuration weakness or credential weakness (such as weak role-based access control policy or API exposure) will allow sensitive healthcare information, governed by HIPAA-like regulation, to get rerouted or exposed.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

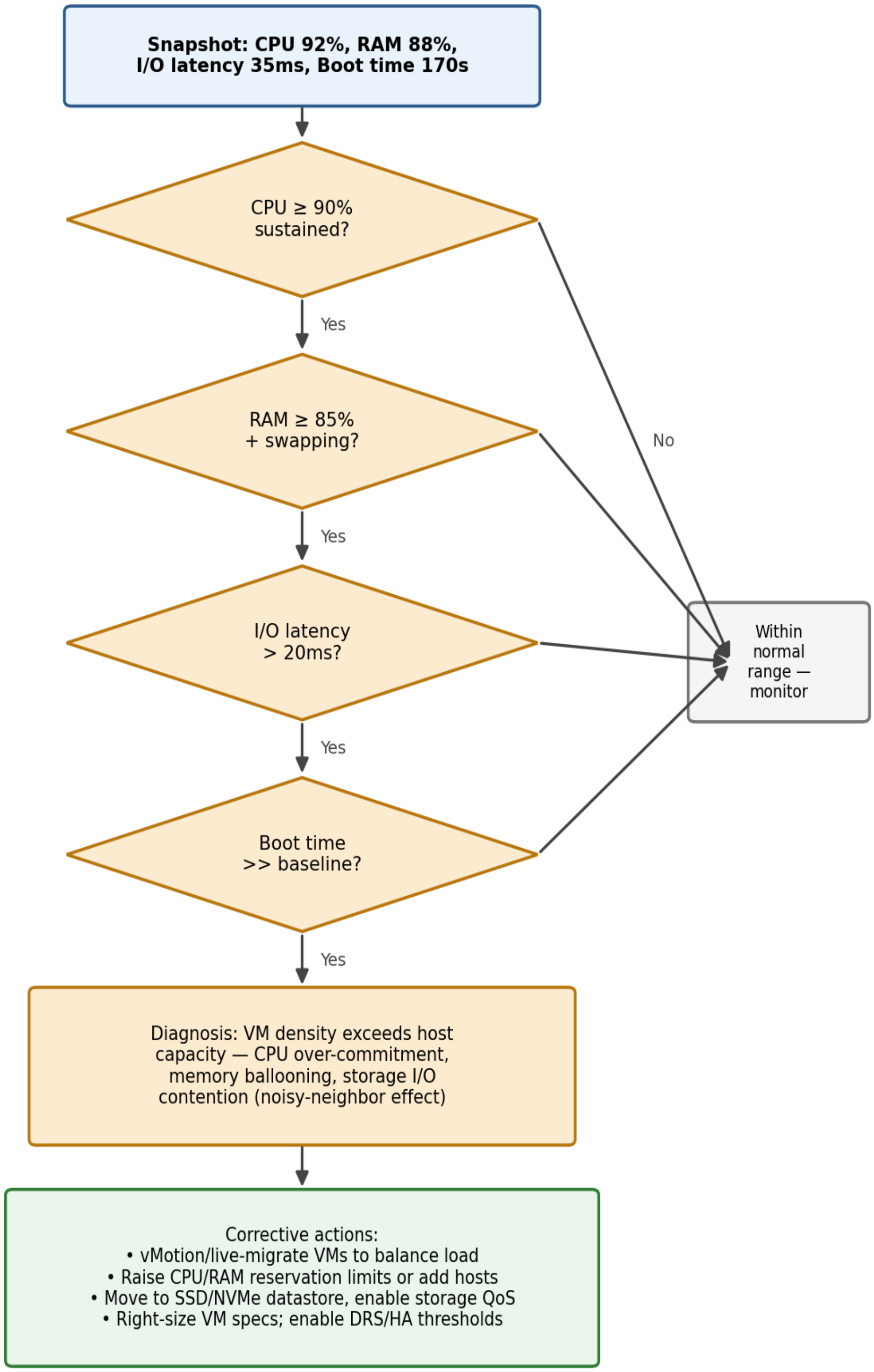
Understanding the snapshot

All four metrics lead in the same direction – the host suffers from over-subscription compared to its hardware capabilities rather than from one particular issue:

- CPU load 92% on average shows CPU over-commitment when too many virtual CPUs are running on too few physical CPU cores resulting in CPU Ready wait time for the VM.
- 88% Memory usage with ballooning/swapping activity at this level demonstrates that the hypervisor takes RAM from VMs which leads to guest-level swap operations and affects response time of applications.
- The latency of 35 ms for storage I/O is clearly above the healthy level of <10-20 ms in case of enterprise-level storage systems. It means that there is probably the problem with datastore where many VMs try to compete for the same queue of the disk.
- The average boot time of 170 seconds (in comparison with normal 30-60 seconds) proves that the diagnosis is correct. Boot storms are extremely resource-consuming processes and they are among the first victims when CPU and I/O are under heavy load.

Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Bottleneck Diagnosis Flow — Host Utilization Snapshot



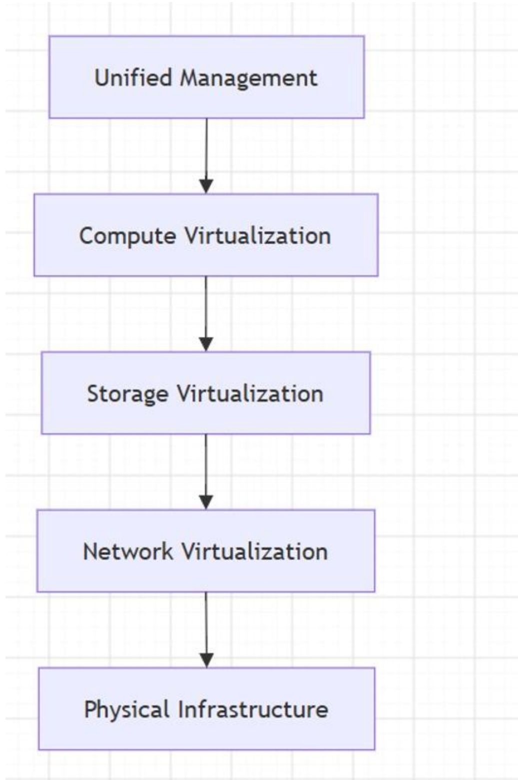
Recommendations for correcting problems

- Balance the load: move VMs using live migration (vMotion/Live Migration) and distribute them over more hosts in the cluster, thus minimizing the workload on each host's CPU and memory.
- Optimize resources: review VMs for vCPU/vRAM over-allocation, minimize oversized VMs, and limit their CPU/memory resources to avoid excessive consumption of the resources.
- Increase capacity: add hosts to the cluster or migrate to a faster storage (SSD/NVMe, or an extra storage with higher IOPS) to restore the I/O latency below threshold.
- Use QoS and DRS/HA features: set up storage I/O control to limit noisy-neighbor VMs and Distributed Resource Scheduler to balance the load proactively.
- Schedule booting: for the VDI/boot storm situation, schedule the process of VM booting and use boot cache to avoid peak I/O requests.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

With the traditional data center architecture, applications are tied down to particular physical servers, storage, and networking hardware, meaning that any time there is a need to make a change – whether to scale up the application, create a new tenant isolation, or even set up a new environment – it needs a lot of hardware-related configuration that may take days or even weeks to complete.

On the other hand, with SDDC, the computation, storage, and networking resources are virtualized to pools that are controlled by software.



Virtualization of compute resources

> The pool of physical servers is made into a hypervisor cluster; VMs (or containers) can be created, duplicated or autoscaled in real time and migrated live between the physical hosts; thus the computing capacity scales according to the workload, rather than vice versa.

Virtualization of storage (Software-Defined Storage)

> Disk capacity in many physical array systems is pooled into flexible volumes, configurable via policies (such as thin provisioning, tiering and replication); therefore, storage capacity and performance tiers can be allocated for a certain workload instantly, without a need to ask a SAN engineer to allocate a new LUN.

Virtualization of networking (SDN / NFV)

> Network functions (switching, routing, firewalling and load balancing) are abstracted from the physical infrastructure and become software overlays (VXLAN, NSX and so on); therefore, a new network segment, a security policy or a firewall rule can be added for a new tenant programmatically for the whole datacenter, without changes in the physical cabling and configuration of hardware-based ACLs.

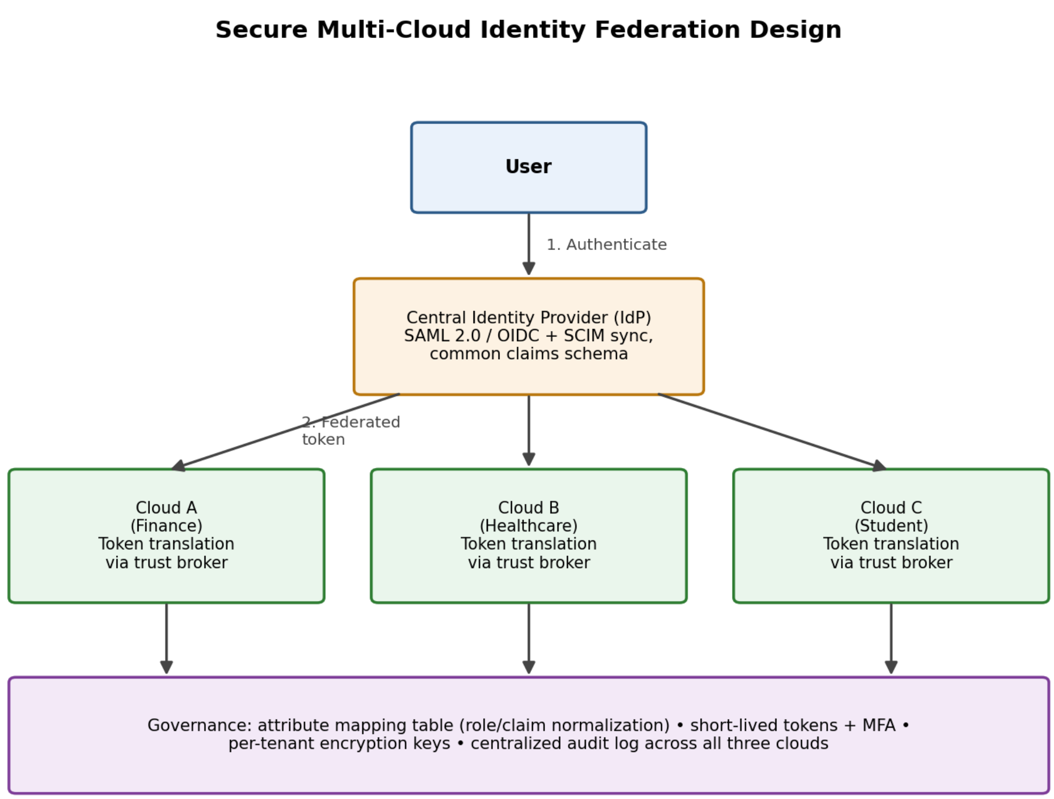
Agility impact

As all the three layers of infrastructure are now controlled via software APIs, SDDC provides self-service provisioning, consistent policy management and elastic scaling, which are the key characteristics of agility.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Root cause analysis

The issue of identity mismatch within federated multi-cloud architecture occurs because each cloud has an identity store, attributes, and tokens that are unique. Due to the absence of any common trust anchor, the user's role or claims from the identity of one cloud are not recognized or mapped appropriately in another cloud, thus causing the failure in the process of authentication and authorization or even granting privileges beyond what the user is supposed to have. Contributing factors include: No common identity provider/trust anchor, different attribute names (role/groups), no common federation protocol, and lack of centralized auditing.



Proposed design for secure identity and privacy

- A central IdP as a trust anchor: implement a single IdP (or federated hub), utilizing the existing standardized protocol such as SAML 2.0 or OpenID Connect/OAuth 2.0, in order to create a trusted entity which will provide authentication credentials to all clouds rather than having separate credential stores on each cloud.
- A common claims/attributes mapping table: define a normalized table for mapping attributes so that the "finance-admin" attribute would have the same meaning and permissions in all clouds avoiding any issues related to inconsistent claim names.
- Lightweight federation broker/token translator per cloud: install an additional service before each cloud which would translate the token provided by the centralized IdP into the format used by that particular cloud. This would allow to easily add/remove any clouds in the future without changing the overall design.
- Least privilege approach, short-lived tokens: provide short-lived, scoped access tokens along with mandatory MFA.
- Protecting data in transit and data at rest: implement identity token and sensitive attribute encryption both in transit (with TLS) and at rest and use per tenant encryption keys for identity data in finance, healthcare, or student clouds.
- Centralized auditing and monitoring: Combine the logs of authentication and authorization activities in all the clouds to one SIEM in order to catch any mismatch, privilege escalation or unusual access across clouds immediately without waiting for any failure to occur.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded: